

Locality of the chiral product forces a nilpotent coalgebra. The OPE has poles of finite order along the diagonal; the logarithmic kernel $\eta_{ij} = d \log(z_i - z_j)$ on the Fulton–MacPherson compactification $\overline{\text{Conf}}_n(X)$ extracts their residues; and the Arnold relation makes $d^2 = 0$ at genus 0. The primitive construction is the ordered bar $B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$: the cofree tensor coalgebra with deconcatenation coproduct, carrying the R -matrix, the KZ associator, and the full Yangian deformation. The factorization coalgebra $\bar{B}(\mathcal{A}) = \text{Sym}^c(s^{-1}\bar{\mathcal{A}})$ on $\text{Ran}(X)$ is its Σ_n -coinvariant quotient, the input to the universal MC element $\Theta_{\mathcal{A}} \in \text{MC}(\mathfrak{g}_{\mathcal{A}}^{\text{mod}})$ governing the genus expansion, the adjunction (Theorem A), and the shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$. At genus $g \geq 1$ the bar differential acquires curvature $d^2 = \kappa(\mathcal{A}) \cdot \omega_g$ from the Hodge bundle; on the Koszul locus, the cobar functor inverts the bar: $\Omega \circ \bar{B} \xrightarrow{\sim} \text{id}$. This chapter builds both objects and establishes the passage between them.

Theorem 1 assembles these three components into the full genus-0 differential d_0 and proves $d_0^2 = 0$; the key input is the Arnold relation.

Information flows in one direction through the chain: from T^c to Sym^c to Lie^c . The symmetrization map $\text{sym}: T^c \twoheadrightarrow \text{Sym}^c$ is a surjective quasi-isomorphism, but its kernel contains the R -matrix’s non-coinvariant Fourier modes: the ordered bar is strictly richer. Theorems A–D use the symmetric bar because their statements are Σ_n -invariant; Part V uses the ordered bar because the quantum-group and Yangian structures it carries are not.

In this chapter, the unadorned notation $\bar{B}^{\text{geom}}(\mathcal{A})$ denotes the symmetric bar $\text{Sym}^c(s^{-1}\bar{\mathcal{A}})$: the coproduct is the coshuffle on Sym^c , matching the geometric decomposition of $\overline{C}_{n+1}(X)$ into unordered collision strata. Every theorem in this chapter is Σ_n -invariant.